
AI 510 Artificial Intelligence of Cloud Computing HOS01A-1 – VS Code and Jupyter Notebooks

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Before You Start

- The directory path shown in the screenshots may be different from yours.
- Some steps are not explained in the tutorial. If you are not sure what to do:
 1. Consult the resources listed below.
 2. If you cannot solve the problem after a few tries, please contact the student worker through the MS Teams course channel.

Learning Outcome

- Set up the working environment for Python
- Understanding the basics VS Code
- Installing VS Code on Mac
- Installing VS Code on Windows
- Installing Jupyter Extension
- Running a simple program in a Jupyter Notebook

Resources

- Jupyter Notebook: An Introduction <https://realpython.com/jupyter-notebook-introduction/>
- The Jupyter Notebook: <https://jupyter-notebook.readthedocs.io/en/stable/>

Install VS Code

Visual Studio Code is a streamlined code editor supporting development operations like debugging, task running, and version control. It aims to provide the tools a developer needs for a quick code-build-debug cycle and leaves more complex workflows to fuller featured IDEs, such as the Visual Studio IDE.

Installation Instruction

1. Please go to the following link and follow the instructions to install Python on your device
<https://cityuseattle.github.io/docs/environment/python/>
2. Please go to the following link and follow the instructions to install VS Code on your device
<https://cityuseattle.github.io/docs/tools/vscode/>

Jupyter Notebooks (formerly IPython Notebooks) are an interactive computing environment that enables users to author files that include Python code, Plots, Narrative Text, Equations, Images, and Videos.

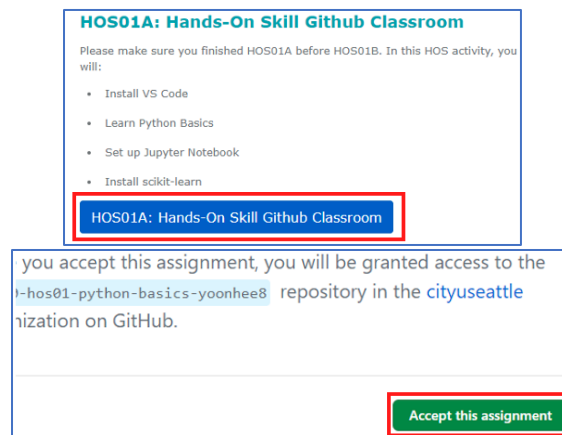
Visual Studio Code (VS Code) supports Jupyter Notebooks.

Environment Set Up

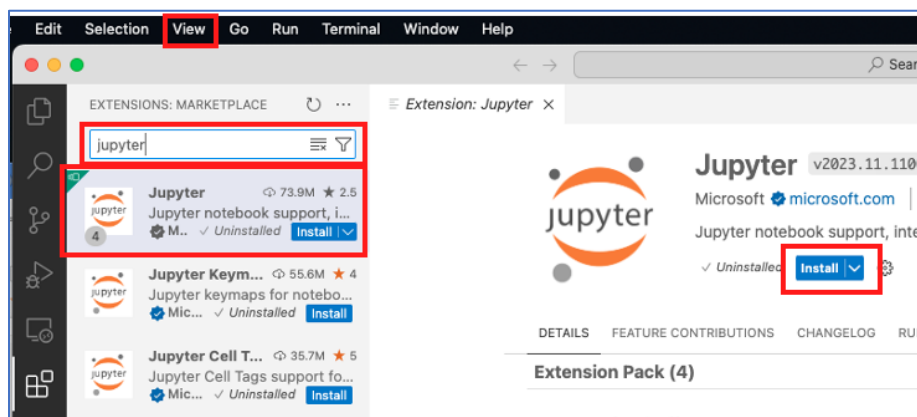
VS Code Extension

1. In Visual Studio Code, open the private repository generated when you accept the HOS assignment. You might not have cloned the repo if you cannot find the repository on your machine. In this case, ensure to do so before proceeding.

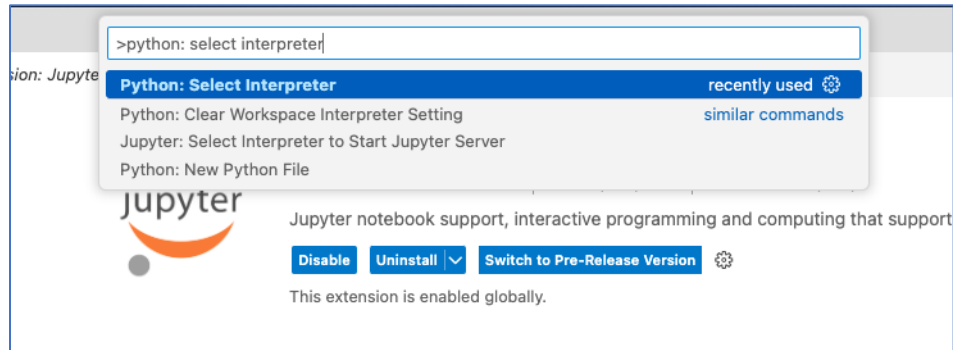
Note: To generate the HOS assignment repository, go to the module in your course shell, click the “GitHub Classroom” button, and click the “Accept this assignment” button as shown in the following screenshot.



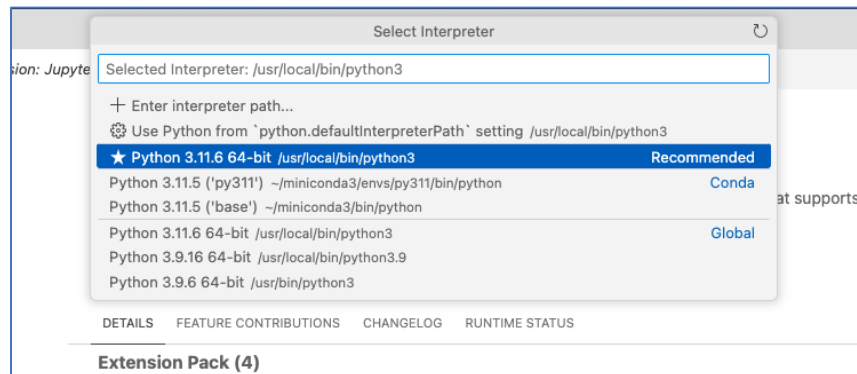
2. From the top menu bar, go to the “View” menu, click the “Extensions” menu, type “Jupyter” in the extension search bar, and click the “Install” button in the Jupyter Notebook item.



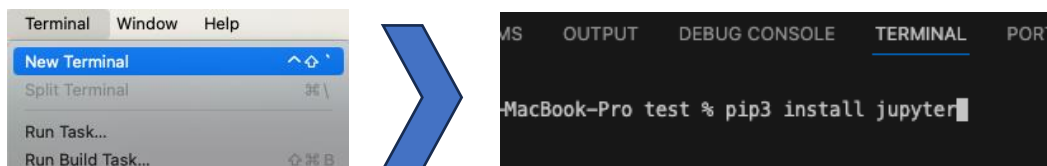
- To select an environment, open **Command Palette** from view or press **Ctrl + Shift ⌘ + P (Windows)** or **Cmd(⌘) + Shift ⌘ + P (OSX)** to bring up a command palette, then type “python: select interpreter” to narrow down the list and then click “**Python: Select Interpreter**”)



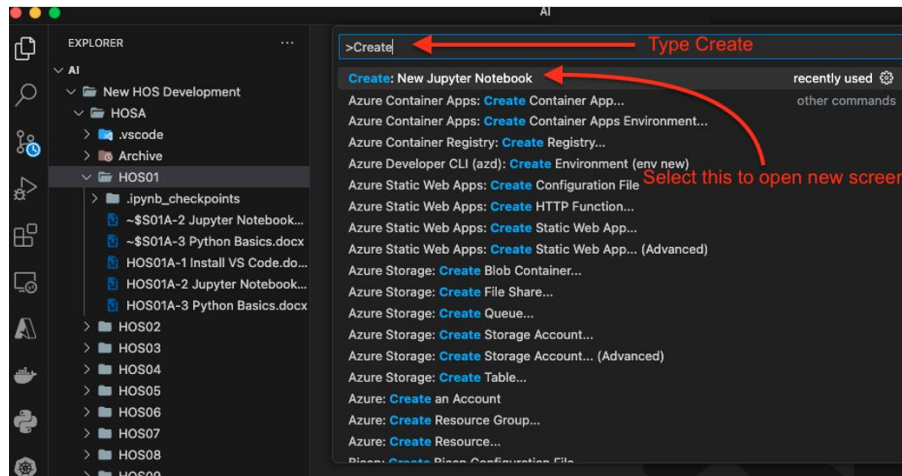
- Choose the latest version (e.g. Python 3.11.6)



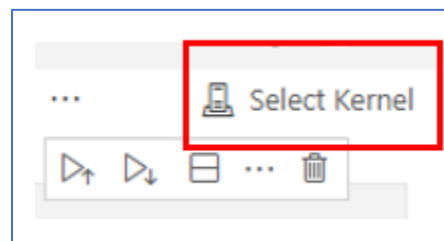
- Open the terminal in VS Code by going to the top “**Terminal**” menu > “**New Terminal**”, and type “**pip3 install jupyter**” to install the Jupyter package.



- Open Command Palette (Press **Ctrl + Shift ⌘ + P (Windows)** or **Cmd(⌘) + Shift ⌘ + P (OSX)**), then type “create” to bring up **Python: Create New Jupyter Notebook** and select it from the dropdown. You will see a screen like the image below.

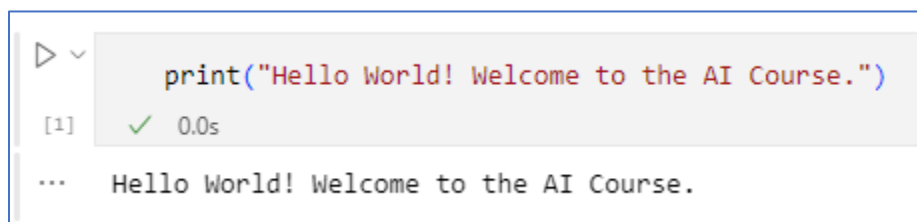


7. When the notebook opens it does not understand which kernel to use nor ipykernel isn't installed yet on the current python environment. We need to select the kernel and install the ipykernel python package. Click the upper right corner of Jupyter Notebook to “**Select Kernel**”.

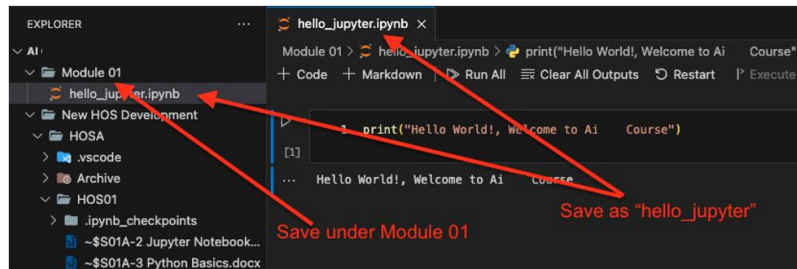


If there is a prompt to install ipykernel, click the **install** button.

8. Type `print(“Hello World! Welcome to the AI Course.”)` into the code writing space under “Untitled-1.ipynb”. After writing the code, click the play button on the left side of the code cell to run the code.



9. Save the file with the name “hello_jupyter.ipynb” under a new folder name “Module 01” (Example: **AI510/Module 01**).



You will notice that the file you saved will have an extension *.ipynb

Take the screenshot for all steps and save it in your local repository.

HOS submission instructions:

1. Please install the GitHub Desktop: https://cityuseattle.github.io/docs/git/github_desktop/
2. Clone, organize, and submit your work through GitHub Desktop: <https://cityuseattle.github.io/docs/hoporhos>